

5(a)	$v = f\lambda$ or $c = f\lambda$	C1
	$f = 3.0 \times 10^8 / 0.040$	C1
	$= 7.5 \times 10^9$ (Hz) $= 7.5$ GHz	A1
5(b)(i)	path difference = 0.020 m	A1
5(b)(ii)	phase difference = 180°	A1
5(c)	(intensity) increases	C1
	(intensity) increases by a factor of 4	A1
5(d)(i)	minimum moves closer to the maximum or decrease in separation of maximum and minimum	B1
5(d)(ii)	the maximum and minimum exchange places or the maximum becomes a minimum and the minimum becomes a maximum	B1